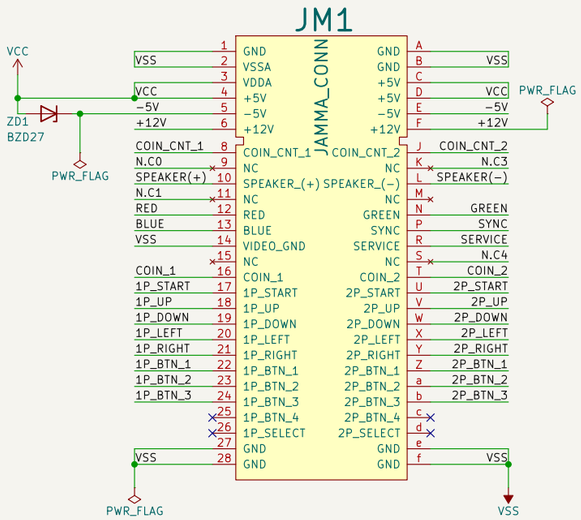
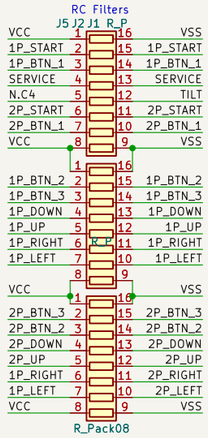
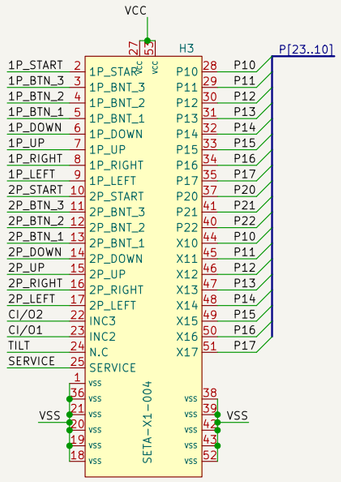
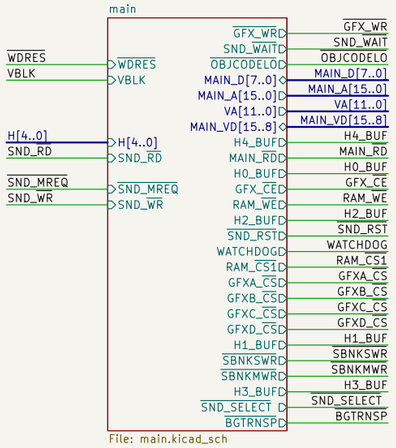
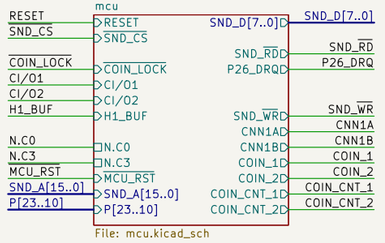


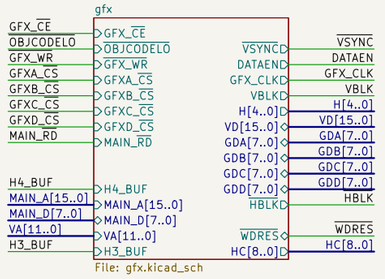
CABINET I/O

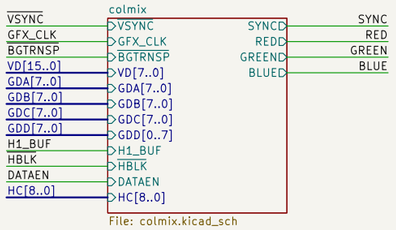


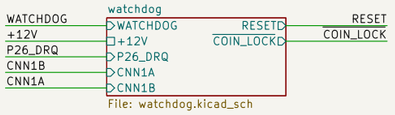




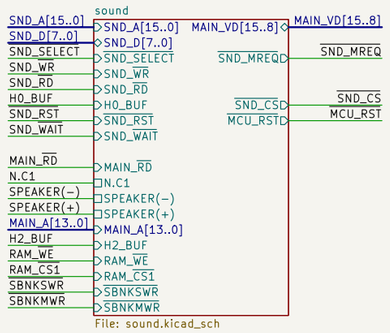






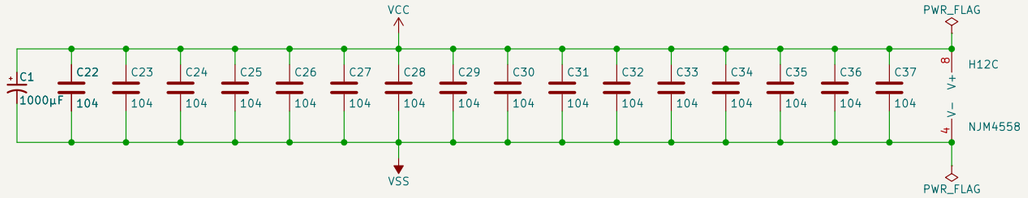






JOTEGO

Sheet: /		
File: exterm.kicad_sch		
Title: EXTERMINATION		
Size: A3	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 1/10



H5E
74LS02

B7C
74LS74

B4C
74LS74

B3C
74LS74

B2E
74LS08

G13G
74LS04

C6E
74LS32

B6E
74LS08

B1G
74LS04

JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: EXTERMINATION

Size: A3	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 2/10

MAIN_A[15..0]

VCC

R330Ω
R19

E8
Z80CPU

WDRES 26 RESET
6 CLK
VCC 17 NMI
16 INT
27 M1
RFSH 28 RFSH
MAIN_WAIT 24 WAIT
18 HALT
MAIN_RD 21 RD
MAIN_WR 22 WR
MREQ 19 MREQ
20 IORQ
25 BUSRQ
BUSACK 23

A0 30 MAIN_A0
A1 31 MAIN_A1
A2 32 MAIN_A2
A3 33 MAIN_A3
A4 34 MAIN_A4
A5 35 MAIN_A5
A6 36 MAIN_A6
A7 37 MAIN_A7
A8 38 MAIN_A8
A9 39 MAIN_A9
A10 40 MAIN_A10
A11 1 MAIN_A11
A12 2 MAIN_A12
A13 3 MAIN_A13
A14 4 MAIN_A14
A15 5 MAIN_A15

D0 14 MAIN_D0
D1 15 MAIN_D1
D2 12 MAIN_D2
D3 8 MAIN_D3
D4 7 MAIN_D4
D5 9 MAIN_D5
D6 10 MAIN_D6
D7 13 MAIN_D7

GND 29
VSS

MAIN_D[7..0]

VCC
B4A
74LS74
VBLKD 2 D
3 CK
5 Q
6 Q̄
1 PR
4 CLR

C9
27C512

MAIN_A0 10 A0
MAIN_A1 9 A1
MAIN_A2 8 A2
MAIN_A3 7 A3
MAIN_A4 6 A4
MAIN_A5 5 A5
MAIN_A6 4 A6
MAIN_A7 3 A7
MAIN_A8 25 A8
MAIN_A9 24 A9
MAIN_A10 21 A10
MAIN_A11 23 A11
MAIN_A12 2 A12
MAIN_A13 26 A13
BANK_CFG0 27 A14
BANK_CFG1 1 A15

VSS 20 CE
RM_OEA 22 OE

D0 11 MAIN_VD8
D1 12 MAIN_VD9
D2 13 MAIN_VD10
D3 15 MAIN_VD11
D4 16 MAIN_VD12
D5 17 MAIN_VD13
D6 18 MAIN_VD14
D7 19 MAIN_VD15

GND 14
VCC
VSS

MAIN_VD[15..8]

MAIN_VD8 3
MAIN_VD9 4
MAIN_VD10 6
MAIN_VD13 11
MAIN_VD12 13
MAIN_VD11 14

74LS174
9 Cp
WDRES 1 Mr
hex flip flop
D0 Q0 2
D1 Q1 5
D2 Q2 7
D3 Q3 10
D4 Q4 12
D5 Q5 15
B5

BANK_CFG[2..0]
BANK_CFG0
BANK_CFG1
BANK_CFG2

SND_RST

WDOGCLR
WDRES
B1C
74LS04
5
6 WATCHDOG

74LS08
B2A
WDRES 1
2
3 SND_RST

D6
PAL16L8

MAIN_A15 1 I1
MAIN_A14 2 I2
MREQ 3 I3
RFSH 4 I4
BANK_CFG2 5 I5
BANK_CFG0 6 I6
VA11 7 I7
OBJCODELO 8 I8
OBJCODEHI 9 I9
H3_BUF 11 I10

VCC 20
GND 10

IO1 19 GFXA_CS
IO3 17 GFXB_CS
IO4 16 GFXC_CS
IO5 15 GFXD_CS
IO6 14 ROMB_A14
IO7 13 RM_OEB
IO8 12 RM_OEA

VA[11..0]

VCC
VSS

R330Ω
R2

B13
74LS245
H[4..0]
H2 2 A0
3 A1
4 A2
H0 5 A3
6 A4
H1 7 A5
H3 8 A6
H4 9 A7
buffers
VCC 1 A->B
VSS 19 CE

B0 18 H2_BUF
B1 17
B2 16
B3 15 H0_BUF
B4 14
B5 13 H1_BUF
B6 12 H3_BUF
B7 11 H4_BUF

B1A
74LS04
1
2

RR15
R150Ω
LOW PASS FILTER

CC4
C101
VSS

BANK_CFG1 9
MAIN_A15 10
B2C
74LS08
8

C11
27C512

MAIN_A0 10 A0
MAIN_A1 9 A1
MAIN_A2 8 A2
MAIN_A3 7 A3
MAIN_A4 6 A4
MAIN_A5 5 A5
MAIN_A6 4 A6
MAIN_A7 3 A7
MAIN_A8 25 A8
MAIN_A9 24 A9
MAIN_A10 21 A10
MAIN_A11 23 A11
MAIN_A12 2 A12
MAIN_A13 26 A13
ROMB_A14 27 A14
ROMB_A15 1 A15

VSS 20 CE
RM_OEB 22 OE

D0 11 MAIN_VD8
D1 12 MAIN_VD9
D2 13 MAIN_VD10
D3 15 MAIN_VD11
D4 16 MAIN_VD12
D5 17 MAIN_VD13
D6 18 MAIN_VD14
D7 19 MAIN_VD15

GND 14
VCC
VSS

D8
PAL16L8

MAIN_A15 1 I1
MAIN_A14 2 I2
MAIN_A13 3 I3
MAIN_A12 4 I4
MAIN_A11 5 I5
MAIN_A10 6 I6
MAIN_A9 7 I7
MREQ 8 I8
RFSH 9 I9
MAIN_WR 11 I10

VCC 20
GND 10

IO1 19
IO2 18
IO3 17 OBJCODEHI
IO4 16 OBJCODELO
IO5 15 WDOGCLR
IO6 14 BGTRNSP
IO7 13 GFX_WR
IO8 12

74LS08
B6B
5
4
6 GFX_CE

R31
R1KΩ
R1
R1KΩ

C3
PAL16L8

MAIN_WR 2 I1
SND_MREQ 3 I2
SND_WR 4 I3
5 I4
6 I5
7 I6
8 I7
SND_CLK' 9 I8
SND_CLK' 11 I9
I10

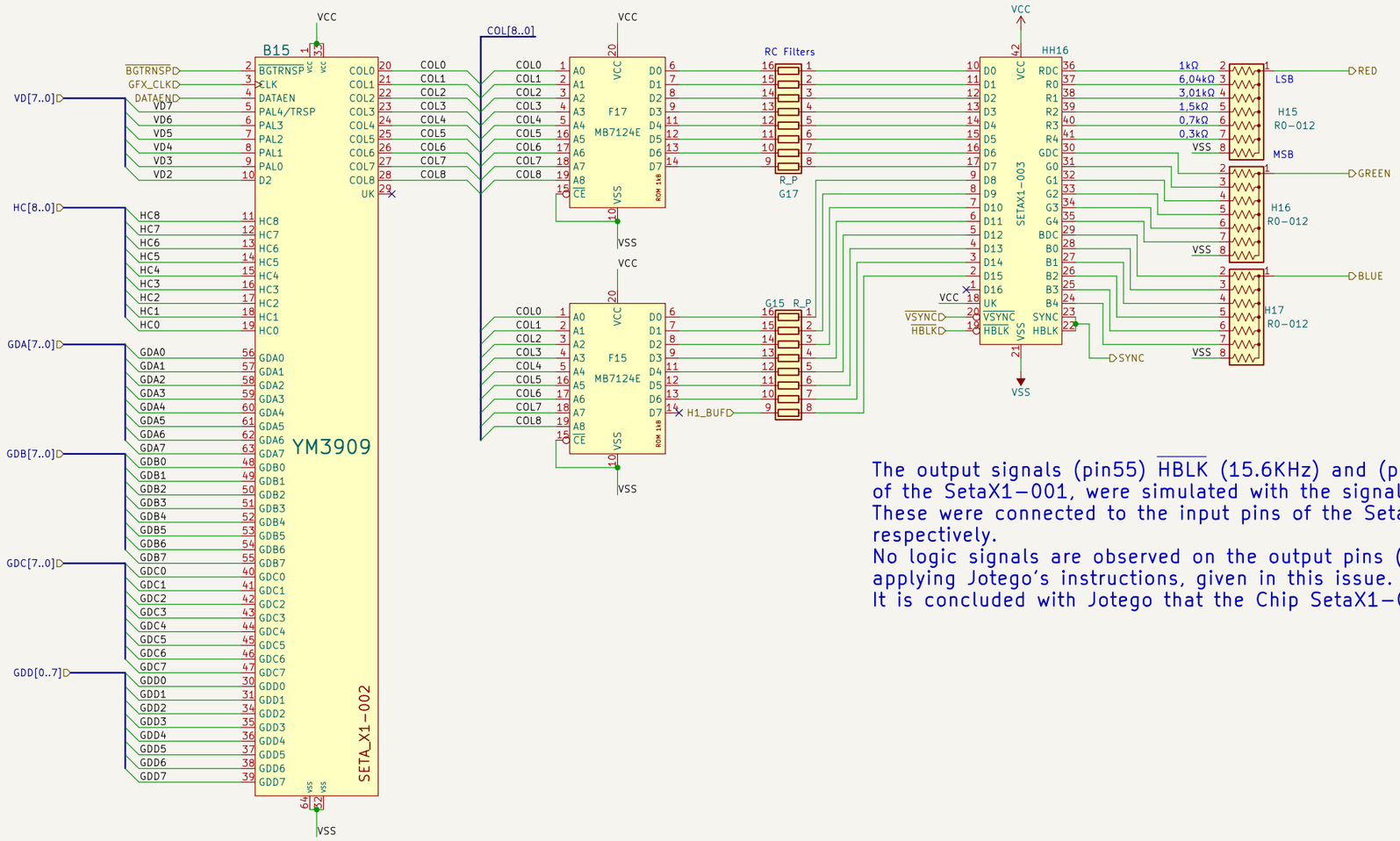
VCC 20
GND 10

IO1 19 SND_SELECT
IO2 18 RAM_WE
IO3 17 RAM_CS1
IO4 16 SBNKMWR
IO5 15 SBNKSWR
IO6 14 MAIN_WAIT
IO7 13 SND_WAIT
IO8 12

H2_BUF 4
H3_BUF 5
B2B
74LS08
6

B4B
74LS74
12 D
11 CK
9 Q
8 Q̄
10 PR
13 CLR

JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: EXTERMINATION
Size: A3 | Date: 29/08/2022 | Rev: José Tejada G.
KiCad E.D.A. 10.0.4 | Id: 3/10



The output signals (pin55) $\overline{\text{HBLK}}$ (15.6KHz) and (pin56) $\overline{\text{VSYNC}}$ (59.2 Hz) of the SetaX1−001, were simulated with the signal generator and an inverter. These were connected to the input pins of the SetaX01−003 (pin 19 − 20), respectively.
No logic signals are observed on the output pins (22−23) HBLK − SYNC, applying Jotego's instructions, given in this issue.
It is concluded with Jotego that the Chip SetaX1−001 is broken.

JOTEGO

Sheet: /colmix/		
File: colmix.kicad_sch		
Title: EXTERMINATION		
Size: A3	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 4/10

GA[15..0]

VD[15..0]

E11

MAIN_D[7..0]

MAIN_D1 2
MAIN_D0 3
MAIN_D7 4
MAIN_D6 5
MAIN_D5 6
MAIN_D4 7
MAIN_D3 8
MAIN_D2 9

A0 A1 A2 A3 A4 A5 A6 A7 — buffers — B0 B1 B2 B3 B4 B5 B6 B7

18 VD9
17 VD8
16 VD15
15 VD14
14 VD13
13 VD12
12 VD11
11 VD10

MAIN_RD 1
A->B
CE 19
74LS245

OBJCODELO 2
GFX_WR 1
74LS08 B6A
3
12 C6D 74LS32 11
13

VD10 3
VD11 4
VD12 7
VD13 8
VD14 13
VD8 14
VD9 17
VD15 18

D0 D1 D2 D3 D4 D5 D6 D7 — A16 — Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
flip flops
74LS273

H4_BUF 11
VCC 1
Cp
Mr

2 GA7
5 GA8
6 GA9
9 GA10
12 GA11
15 GA5
16 GA6
19 GA12

E12

MAIN_D1 2
MAIN_D0 3
MAIN_D7 4
MAIN_D6 5
MAIN_D5 6
MAIN_D4 7
MAIN_D3 8
MAIN_D2 9

A0 A1 A2 A3 A4 A5 A6 A7 — buffers — B0 B1 B2 B3 B4 B5 B6 B7

18 VD1
17 VD0
16 VD7
15 VD6
14 VD5
13 VD4
12 VD3
11 VD2

MAIN_RD 1
A->B
CE 19
74LS245

H3_BUF
2 74LS32 C6A 3
1

VD1 3
VD0 4
VD5 6
VD4 11
VD3 13
VD2 14

D0 D1 D2 D3 D4 D5 — Q0 Q1 Q2 Q3 Q4 Q5
74LS174
hex flip flop

H4_BUF 9
VCC 1
Cp
Mr A15

2 GA14
5 GA13
7
10 ROM_CE
12 GA15
15 ROM_OE

VCC
R4 3.3kΩ
R3 22Ω
A17
Standby VDD GND Out
1 3 4 2
12.000MHz
VSS

VCC
VCC

C17

MAIN_RD 2
GFX_WR 3
RD
WR

63 CLK
VD7 60
VD6 59
VD0 4
FLIPX
FLIPY
XC0

VD15 7
VD14 8
VD13 9
VD12 10
VD11 11
VD10 12
VD9 13
VD8 14
D7 D6 D5 D4 D3 D2 D1 D0

VA[11..0]
VA0 15
VA1 16
VA2 17
VA3 18
VA4 19
VA5 20
VA6 21
VA7 22
VA8 23
VA9 24
VA10 25
VA11 28
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11

SPBANK1 26
MAIN_A11 27
SPBANK1
SPBANK2

MAIN_A[15..0]

5 UK
6 UK
31 UK
45 UK
44 UK

VSS 64 VSS 32

YM3906
SETA_X1-001

CLK 62 GFX_CLK
VBLK 58
HBLK2 57
VSYNC 56 VSYNC
HBLK 55 HBLK
DATAEN 61 DATAEN

HC8 64 HC8
HC7 53 HC7
HC6 52 HC6
HC5 51 HC5
HC4 50 HC4
HC3 49 HC3
HC2 48 HC2
HC1 47 HC1
HC0 46 HC0

HC[8..0]

H8 43 H4
H4 42 H3
H2 41 H2
HPH90 40 H0
H 39 H1

H[4..0]

CHA0 35 GA0
CHA1 36 GA1
CHA2 37 GA2
CHA4 38 GA4
H8B 34 HR8
~{WDRES} 29 WDRES
RW 30

B1F 74LS04 13 12 VBLK

MAIN_RD 9 B1D 74LS04 8
74LS32 C6B 5 4 6 GFX_WR

Measures Seta X1-001:
CLK_IN.....12MHz
CLK_OUT.....11.07-12.01MHz
VBLK.....59.2Hz
VBLK.....59.3Hz
HBLK2.....15.6KHz
HBLK.....15.6KHz
VSYNC.....59.2Hz

Observatión: Pin 46 (HC0) does not
correspond to the sequence of the HC[8..0] counter.

gfx_ram

H3_BUF
MAIN_A[15..0]
GFXA_CS
GFXB_CS
GFXC_CS
GFXD_CS
GFX_WR

SPBANK1
VA[11..0]
VD[15..0]

File: gfx_ram.kicad_sch

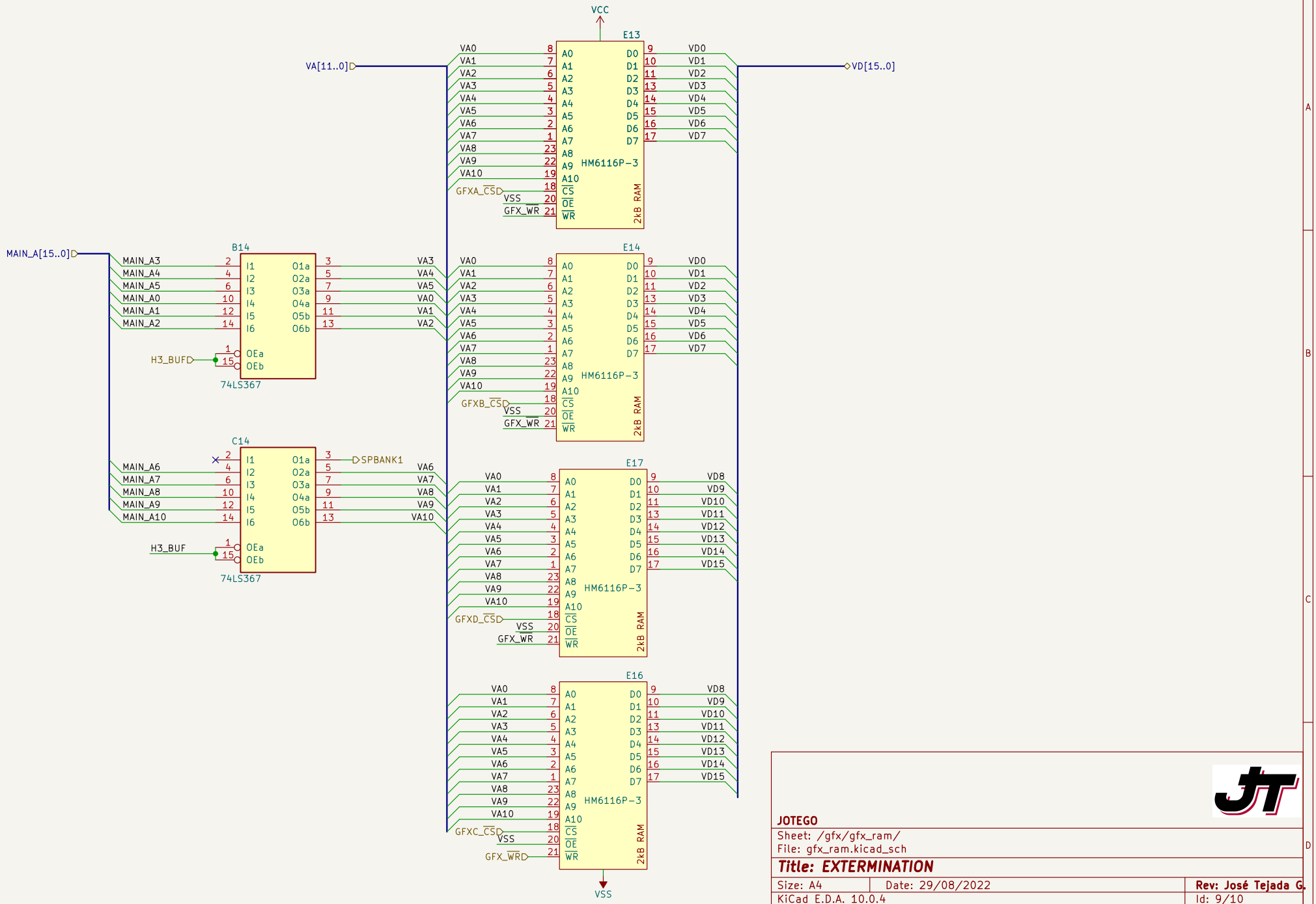
gfx_rom

ROM_CE
ROM_OE
GA[15..0]
HR8

GDA[7..0]
GDB[7..0]
GDC[7..0]
GDD[7..0]

File: gfx_rom.kicad_sch

JOTEGO
Sheet: /gfx/
File: gfx.kicad_sch
Title: EXTERMINATION
Size: A3 | Date: 29/08/2022 | Rev: José Tejada G.
KiCad E.D.A. 10.0.4 | Id: 5/10



JOTEGO

Sheet: /gfx/gfx_ram/
File: gfx_ram.kicad_sch

Title: EXTERMINATION

Size: A4	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 9/10

NOT POPULATED

GA[15..0]

HR8

ROM_CE

ROM_OE

VCC

VSS

GND

27C512

A13

A12

A10

A8

A7

A5

A4

A2

GDA[7..0]

GDB[7..0]

GDC[7..0]

GDD[7..0]

ROM_CE 11 B1E 10 ROM_CE'

74LS04

JOTEGO

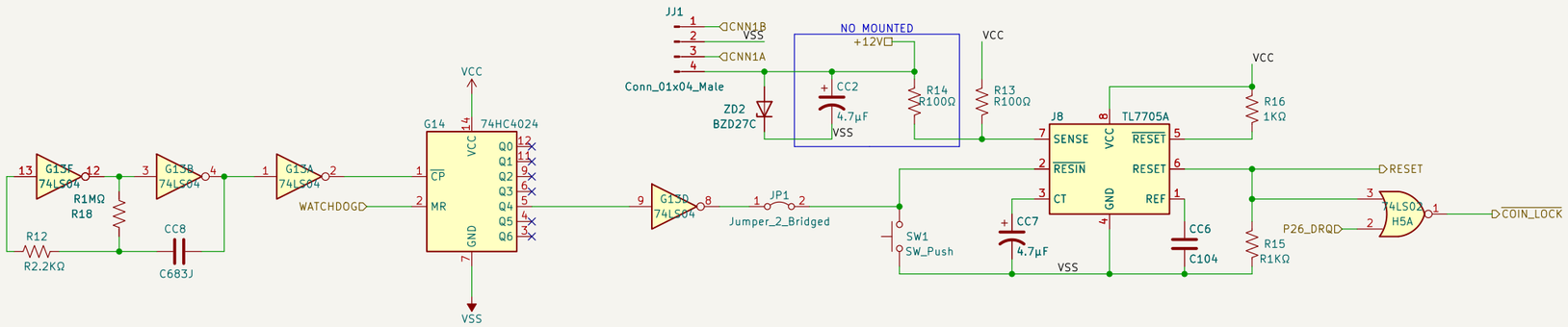
Sheet: /gfx/gfx_rom/

File: gfx_rom.kicad_sch

Title: EXTERMINATION

Size: A3 | Date: 29/08/2022 | Rev: José Tejada G.

KiCad E.D.A. 10.0.4 | Id: 10/10

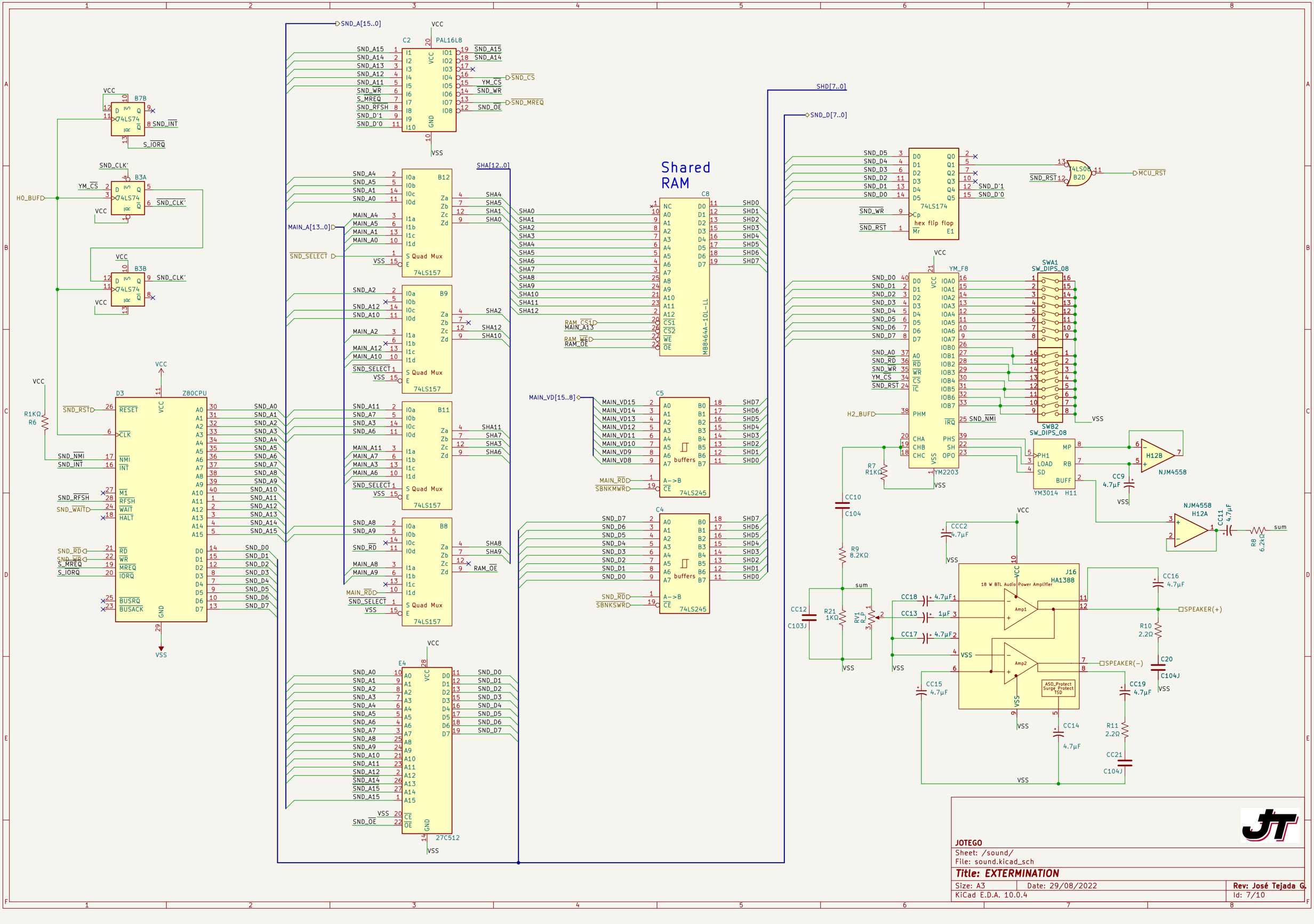


JOTEGO

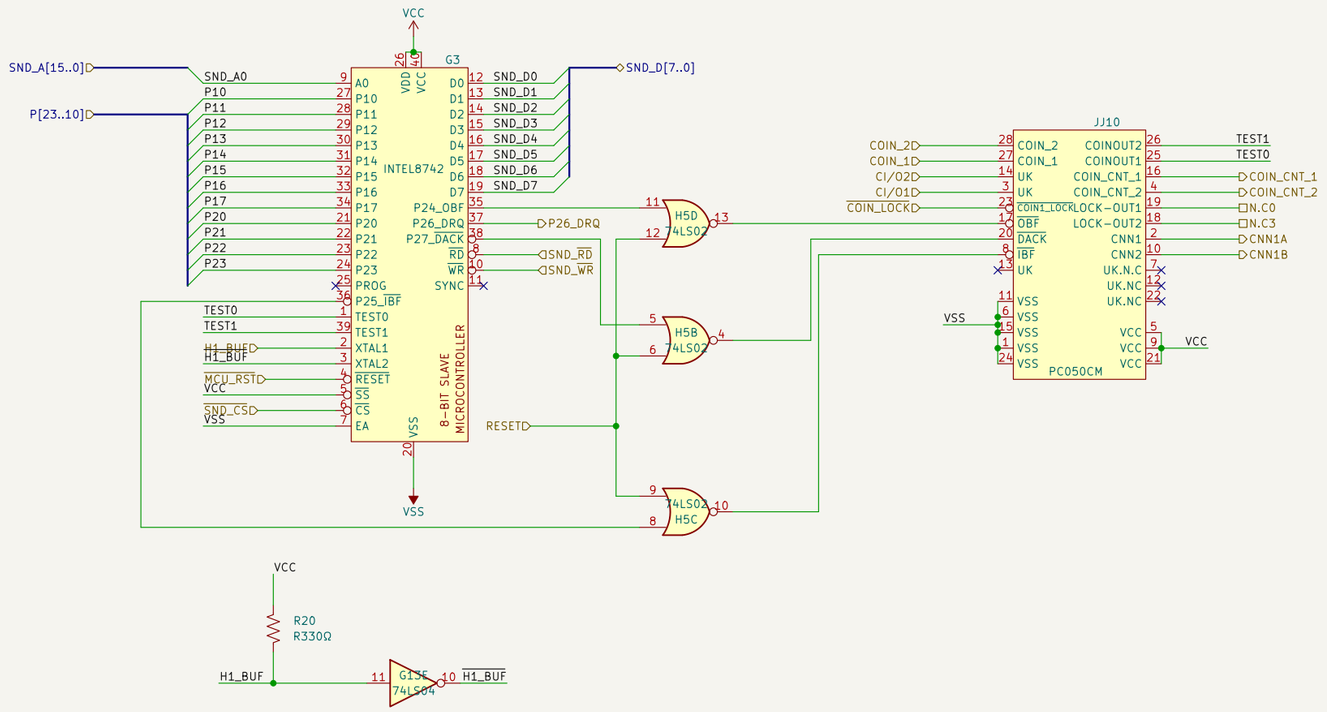
Sheet: /watchdog/
File: watchdog.kicad_sch

Title: EXTERMINATION

Size: A3	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 6/10



JOTEGO	
Sheet: /sound/	
File: sound.kicad_sch	
Title: EXTERMINATION	
Size: A3	Date: 29/08/2022
KiCad E.D.A. 10.0.4	Id: 7/10
	Rev: José Tejada G.



JOTEGO		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: EXTERMINATION		
Size: A3	Date: 29/08/2022	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 8/10